



Spreading life between planets

Research suggests that clumps of bacteria could spread life around the universe, like pollen. <http://digit.in/sep20-63>



Where did water come from?

Meteorites hint that water may have arrived along with the rocks that formed the planet. <http://digit.in/sep20-64>

A Neanderthal skull in the foreground, with a Homo Sapien skull in the background

WHY DO HUMANS HAVE CHINS?

An enduring morphological mystery

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he chin, also known as the mental eminence, or the mentum osseum is one of the few parts

of the skeleton that can identify an anatomically modern human from other remains. If a skull is found, the first thing scientists do is to check if it has a chin. This approach has not been without controversy of its own. This is because there are some finds that are on the borderline. Depending on the way you look at it, either some Neanderthals had chins, or those remains are not of Neanderthals. There are also some specimens that are the result of interbreeding between Neanderthals and Sapiens. If the remains have a chin, that means that it is human.

For over a century, scientists have proposed different ideas on why we might have evolved chins. In 2016, James Pumpush and David Daegling wrote a review titled “The Enduring Puzzle of the Human Chin” that looked at all the different theories and their relative weak or strong points.

There are two main schools of thought when it comes to explaining a chin. The first is that it is a spandrel, a leftover structure because of all the other evolutionary changes that took place on the face. The word spandrel comes from decorative corners of walls that were filled with ornate and decorative elements, just to fill the space, but were later proved to be useful for additional support. The term was adapted for biology from architecture in a 1979 paper, “The spandrels of San Marco and the Panglossian paradigm: a critique of the adaptationist programme by Stephen Gould and Richard Lewontin.” In it, the authors say that not everything in a species is an adaptation dictated by the forces of evolution, and the

main example cited by the paper was the human brain! Let that not bother you, other examples include the overly masculinised penises of female hyenas, and the brooding chambers of some snails. In the context of the evolution of the human chin, it might just be a feature that did not evolve over a series of generations because it was advantageous, just that it was allowed to stay because it was not disadvantageous. This means that the genes generally drifted towards creating a chin, which should not be possible considering the relatively short time window when the chin emerged as compared to the predecessors of the Sapiens. Which brings us to the other possibility, that the chin is not a spandrel at all, but an adaptation that actually has some purpose. This is suspected to be something to do with the structure of the jaw and the facial muscles, as well as the way the teeth of Sapiens evolved, but exactly what, we cannot pinpoint.

One of the oldest theories was that the teeth of the humans became